RESEARCH REPORT · VITKI

Fully tested finding

Sons and brothers died outside Scandinavia more often than fathers and husbands – 10 against 3 percent

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Report version 3 · Published 31 October 2026 · Analysis in Vitki 3.0.0 · Licence CC BY 4.0

Last changed 5 November 2026

Timeline of the study: The analysis plan was locked 28 October 2026 · The material was frozen 29 October 2026

ABSTRACT

Background and aim. The runestones for men who died in the east and the west are among the best known, and the words for the dead – *drengr* and *þegn* – have been read as words for young and older men respectively. The question here is whether those who died outside Scandinavia were more often sons and brothers than fathers and husbands, with time, region and carver held constant.
Material. 945 legible Swedish Viking-Age runestones on which all the dead are men and the dead man is either a son or brother (a younger place in the family) or a father or husband (an older place), from the Scandinavian Runic-text Database (Rundata): 393 and 552 stones respectively.
Method. Pre-registered logistic regression with family place, region and dating period as explanatory variables and the carver as a random effect; Holm correction over the study's six tests; internal replication in two pre-defined halves; eight sensitivity analyses; adjustment for all 48 associations the project reports.
Results. The stone says the dead man died outside Scandinavia on 9.7 percent of the stones for sons and brothers against 2.6 percent for fathers and husbands; odds ratio 4.10 (95 % interval 2.18–7.69), $p < 0.001$, Holm-corrected $p < 0.001$. Same direction and significant in both halves, and every sensitivity analysis holds, also without one province at a time; adjusted for all associations $q < 0.001$. Of the 69 stones for men who died outside Scandinavia, 4 (6 percent) were raised by a wife, against 117 of 1,148 (10 percent) of the other stones for men – a difference that may be chance ($p = 0.30$).
Conclusion. Death in the east and the west struck first of all men whom the stones remember as sons and brothers, not as fathers and husbands. That fits the reading of *drengr* as a word for the young man, but the measure is a place in the family, not an age, and it concerns only those who died abroad.

Keywords: *runestones, Viking Age, journeys abroad, kinship, life stage, drengr, þegn, logistic regression, pre-registration*

HIGHLIGHTS

- Those who died in the east and the west were more often sons and brothers than fathers and husbands – three to four times as often, and the difference withstands every check in the study.
- On the stones for those who died abroad it is most often parents and siblings who remember them, more seldom a wife and children.
- Whether the journeys themselves belonged to a stage of life before a family of one's own the stones cannot say: they concern only those who never came home.

1. Introduction

Runestones for men who died abroad are among the most studied. Mats G. Larsson (1990) went through the stones about journeys abroad and what they say about the society behind them. Judith Jesch (2001) examined the vocabulary of runic inscriptions and skaldic verse on journeys and retinues, among them the words *drengr* and *þegn*. Lauren Goetting (2006) argued that *drengr* was used chiefly of young men in a warrior retinue and *þegn* of older, established men, and Denis Sukhino-Khomenko (2024) followed *þegn* around the North Sea. None of them measured whether the man who died abroad was more often a young man in his family than an older one, with time, district and carver held constant.

The aim of this study is to test that, with the family place as the measure: a son or brother stands in a younger place in the family than a father or husband. That is not the same as age, but it is the nearest measure the inscriptions themselves give.

2. Pre-registration: hypothesis and plan

The hypothesis, the model and the rules of interpretation were written down in an analysis plan on 28 October 2026, before any model had been run. The plan concerned a study of who raised the runestones and for whom, with six tests; this is test 5. The plan is time-stamped in the project's version control and has the SHA-256 checksum `ea26a388379e29eb…` (see Data and code).

The test reads in the plan: the outcome is that the stone says the dead man died outside Scandinavia, among stones where the message can be read; the dead man is *young* in the family when he is a child or sibling of the sponsor and neither parent nor spouse, and *older* when he is a parent or spouse and neither child nor sibling; only stones on which all the dead are men; the model is outcome ~ young + region + dating period + carver as a random effect. The six tests are corrected together by Holm's method. A test *confirms* when Holm p is below 0.05 and the direction is the same with p below 0.05 in each of the material's two halves; a *pattern to test* has Holm p below 0.05 but does not hold in both halves.

Before the plan was locked no count of death abroad by family place had been made. For tests 3 and 4 on *drengr* and *þegn*, however, I had seen a rough count without a model: *drengr* on 8 percent of the stones for sons and brothers against 0.4 percent for fathers, and *þegn* the other way round. The plan says so. Tests 3 and 4 are therefore reported in section 5.5 as complementary, not as main results.

Sections 5.1–5.3 are the analyses the plan prescribes for test 5. Section 5.4 is descriptive counts on the same selection, made after the test. No other analyses are included.

3. Material

- **Source.** The Scandinavian Runic-text Database (Rundata), Department of Scandinavian Languages, Uppsala University: transliteration, normalisation, English translation, province, parish, carver and style group for each inscription. The material was frozen for the analysis on 29 October 2026.
- **The stones.** Viking-Age runestones in the Swedish provinces: 2,320 stones. Runestones only; other objects with runes are not included. Skåne, Blekinge and Halland, Danish in the Viking Age, are counted as Denmark in Rundata and are not included.
- **Selection.** Stones whose text is preserved well enough for the dead man and his relation to the sponsor to be read, on which all the dead are men, and on which the dead man is either young or older in the family by the plan: 393 stones for sons and brothers and 552 for fathers and husbands, 945 in all.
- **The family place.** The relation is read from Rundata's English translation by rules (son, brother, father, husband and their forms). The rules were checked against hand coding on a new sample of 40 stones they had never seen: the relation was right on 38 of 40.
- **The outcome** is that the stone says the dead man died outside Scandinavia: a word for death or a manner of dying near a place outside Scandinavia (east, west, Greece, England, Serkland, Ingvar's expedition and others). Having merely travelled does not count, nor dying on a journey at home. The rule was checked against hand coding of 205 stones: the rule's yes was right 92 times of 92 and its no 113 times of 113. The hand coding was partly done while the rules were being corrected and is not an independent test; of 33 randomly drawn stones, 2 of which say the dead man died abroad, the rule was right on all 33.
- **Region** by today's provinces: Svealand (Uppland, Södermanland, Västmanland, Närke and Gästrikland), Götaland (Östergötland, Västergötland, Småland and Öland), Gotland and Norrland (Hälsingland and Medelpad), the provinces with stones in the selection. **Dating period** by the stone's ornamental style (Gräslund 2006): before 1010, 1010–1040, 1040–1070, after 1070, and undated as a group of its own.
- **The carver.** The carver Rundata gives. Stones without a known carver are grouped by parish.

4. Method

The statistical terms are explained in the reports' glossary: https://vitkirunor.se/en/forskningsrapporter/begrepp.

4.1 The model. Logistic regression with the R package mgcv (R 4.6.1; Wood 2017): the outcome against family place, region and dating period, with the carver as a random effect, so that a prolific carver's way of wording the stones is not counted as a pattern. The shares are the model's marginal effects.

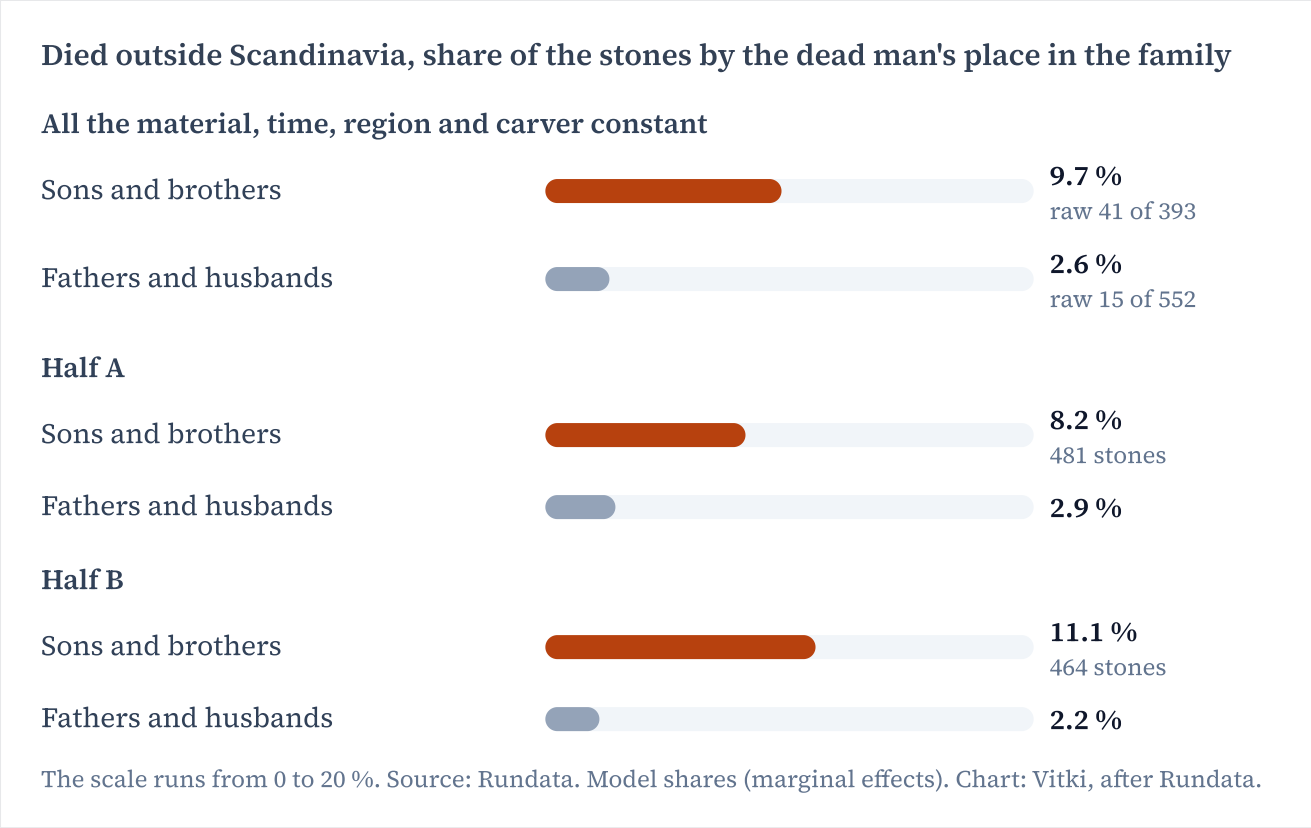
4.2 Correction and replication. Holm correction (Holm 1979) over the study's six tests. Internal replication: the test must hold in each of the material's two halves. The halves are formed with the same rule, fixed before the study, in all the project's analyses (whole parishes dealt alternately within each province in order of the parish's mean dating); since the rule is applied to the selection being tested, the size of the halves differs slightly between the tests. The plan's robustness checks: only stones with at least ten legible words, the model without the carver effect, Uppland only, and only sons against only fathers.

4.3 Sensitivity analyses. The same fixed series as for every association the project reports: the raw model; adjusted for province and dating period; the carver as a random effect; the parish as a random effect; the stone's location as a spatial surface (only stones that according to Rundata stand in place); each half on its own; and without one province at a time (provinces with at least 30 stones). An analysis holds when p is below 0.05 and the direction is the same as in the adjusted model; without a province, significance is required only where at least two thirds of the stones remain. The p-values are also adjusted together for all 48 associations reported (Benjamini & Hochberg 1995); the adjusted value is called q. The model's assumptions are checked with simulated residuals (DHARMa; Hartig 2022): the model simulates new outcomes for each stone many times, and each real outcome is compared with the simulated ones. The checks look for an uneven spread, more variation than the model allows for (overdispersion) and neighbouring stones that are more alike than the model explains (remaining spatial pattern). The errors in the outcome rule are tested by 200 refits in which each stone's ”true” outcome is drawn from the hand-coded sample's hit rates.

4.4 Deviations from the plan. None. The plan's check ”sex determined for every sponsor” applies only to the tests about the sponsors and is not made here.

5. Results

5.1 The main test. Raw, the stone says the dead man died outside Scandinavia on 41 of 393 stones for sons and brothers (10.4 percent) and on 15 of 552 for fathers and husbands (2.7 percent). With time, region and carver constant the shares are 9.7 against 2.6 percent. The odds ratio is 4.10 (95 % interval 2.18–7.69), $p < 0.001$, Holm-corrected $p < 0.001$. In plain words: on a stone for a son or brother it is three to four times as common that the dead man died abroad as on a stone for a father or husband. Half A: 481 stones, odds ratio 3.03 (1.25–7.35), $p = 0.01$. Half B: 464 stones, odds ratio 5.82 (2.30–14.71), $p < 0.001$. The test *confirms* by the plan's rule.



5.2 The plan's robustness checks.

- Only stones with at least ten legible words: 661 stones, odds ratio 4.80, $p < 0.001$.
- Without the carver effect: odds ratio 4.02, $p < 0.001$.
- Uppland only: 489 stones, odds ratio 4.93, $p = 0.003$.
- Only sons against only fathers: 653 stones, odds ratio 5.26, $p < 0.001$. Raw 26 of 204 against 12 of 449.

5.3 The sensitivity analyses (945 stones, 56 with the outcome). All 8 hold:

- Raw model: odds ratio 4.17 (2.27–7.65), $p < 0.001$.
- Adjusted for province and dating period: 3.90 (2.09–7.27), $p < 0.001$.
- Carver as a random effect (394 carvers and parish groups): 4.41 (2.08–9.34), $p < 0.001$.
- Parish as a random effect (399 parishes): 4.04 (1.98–8.22), $p < 0.001$.
- The stone's location as a spatial surface (567 stones that according to Rundata stand in place): 2.56 (1.19–5.48), $p = 0.02$. The difference is smaller here, but it remains when the district is held constant.
- Half A (476 stones): 3.19, $p = 0.01$. Half B (469 stones): 5.53, $p < 0.001$.
- Without one province at a time (5 provinces tested): the direction holds without each of them and is significant everywhere. Weakest without Uppland, when 456 stones remain: odds ratio 3.64 (1.63–8.11), $p = 0.002$.
- The difference in percentage points in the adjusted model: 9.9 against 2.9 percent, 7.0 points (probably between 3.8 and 10.2).

- The model's assumptions: residual distribution p = 0.56, remaining spatial pattern p = 0.17, but overdispersion p = 0.032: the model does not capture all the variation between the stones, so the intervals are probably somewhat too narrow (see 6).
- Errors in the outcome rule: the association holds in 100 percent of the 200 refits; median odds ratio 3.65, range 2.37–4.17.
- Adjusted for all 48 associations the project reports: q < 0.001. The association is one of 15 that hold in every sensitivity analysis.

5.4 Who they were (descriptive, same selection). Of the 41 sons and brothers who died outside Scandinavia, 26 were sons and 15 brothers. On 13 of the stones a woman stands among the sponsors, and all 13 are mothers commemorating their sons. Of the 69 stones for men who died outside Scandinavia, whatever their relation to the sponsor, 4 (6 percent) were raised by a wife, against 117 of 1,148 (10 percent) of the other stones for men – a difference that may be chance (Fisher's exact test, which computes how often a difference at least this large arises by chance with the same numbers of stones: p = 0.30). Parents or siblings stand among the sponsors on 46 of the 69 stones, children or a wife on 20. The difference between sons and brothers and fathers and husbands is present in both large regions: in Svealand 26 of 261 for sons and brothers against 12 of 423 for fathers and husbands, in Götaland 15 of 124 against 3 of 121.

5.5 The words for the dead (the plan's tests 3 and 4). *drengr* appears on 27 of 393 stones for sons and brothers against 4 of 552 for fathers and husbands: odds ratio 8.7, Holm-corrected p < 0.001, but too few cases in half B to test there. *þegn* appears on 5 of 393 against 26 of 552: odds ratio 0.22, Holm-corrected p = 0.01, but not significant in either half. Both are *patterns to test* by the plan's rule, and for both a rough count was seen before locking.

6. Discussion

What the finding says. The men whose runestones say they died in the east or the west were more often sons and brothers than fathers and husbands. The difference is large, three to four times, and it remains with time, region and carver constant, in each half, without any one province and with the district held constant. Death in the east and the west struck first of all men whom the stones remember as sons and brothers, not as fathers and husbands. That few of their stones were raised by a wife, on the other hand, does not differ clearly from other stones for men and may be chance. Whether the journeys themselves belonged to a stage of life, the time before the man had a family of his own, the measure does not say, since it concerns only those who died abroad.

Against earlier research. The result gives a measure to the reading of the words for the dead that Goetting (2006) argued for: *drengr* for young men, *þegn* for older ones. Here both words follow the family place in the same direction as death abroad, but only as patterns to test. The finding on death abroad rests not on the words but on the relation and is therefore a measure of its own, independent of them: death in the east and the west struck first of all men in a younger place in the family, sons and brothers (cf. Jesch 2001; Larsson 1990). It says nothing about who travelled and came home.

Weaknesses, in order of weight.

1. **Family place is not age.** A son may be middle-aged, a brother may be older than the sponsor. The measure says that the dead man was not yet the one whom others commemorated as father or husband, not how old he was.
2. **Who raises the stone and what is written go together.** A father who died abroad may have been commemorated by sons who worded the stone differently, without a place for the death. The finding concerns what the stones say, and it may partly be a difference in how different sponsors phrased themselves.
3. **Overdispersion.** The model check shows more variation between the stones than the model allows for (p = 0.032). The odds ratio is hardly affected, but the intervals are probably somewhat too narrow. With 56 stones with the outcome they are already wide.
4. **The relation is read by rules.** The rules had 38 right of 40 on a new sample. Errors in the relation are not part of the refits, only errors in the outcome.

What would overturn the finding. A test in which the sponsor is held constant – for example only stones a parent raises for a son compared with stones a son raises for a parent – in which the difference disappeared, or a hand coding of the relation showing more errors among the stones about death abroad.

7. Conclusion

On the runestones for sons and brothers it says more often that the dead man died outside Scandinavia than on the stones for fathers and husbands, 9.7 against 2.6 percent with time, region and carver constant, and the difference withstands every sensitivity analysis. Death in the east and the west struck first of all men whom the stones remember as sons and brothers, not as fathers and husbands. The next step is to hold the sponsor constant and to test the words for the dead on a larger material.

Data and code

The analysis was made in Vitki 3.0.0, an open research tool for runestones (vitkirunor.se). The analysis code as it stood when the results were computed is commit `5542b59` (29 October 2026) in https://github.com/Precatio/runes, the last change to the analysis and its results before report version 1; it is Vitki 3.0.0 with later changes. All code, the analysis plan and the result files are openly available there under GPL-3.0.

- The analysis plan: `analysplaner/2026-10-28-manniskorna.md`, with its checksum in the `.sha256` file beside it.
- The family place and the sponsors: `src/people.py`; the hand coding: `data/people_check.json`.
- The outcome: `src/messages.py` (death outside Scandinavia); the hand coding: `data/messages_facit_ost.csv` and `utdata/brittiska/hemma_kodning.csv`, the agreement per sample in `data/messages_check.json`.

- The model: `r/people.R`. The sensitivity analyses: `r/robustness.R` and `src/robustness.py` (method description in `METHODS.md`, sections 1u and 1v). The results are saved under the material's version `e83202e01a5e6d27`.
- To rerun the analysis: check out commit `5542b59`, start the analysis engine with `scripts/dev_api.sh` and run `curl -X POST http://localhost:8000/api/r/corpus/run`. That computes all the corpus analyses and the sensitivity analyses for the Rundata loaded and saves them under `data/cache/r/corpus/<version>/` (each R module is run as `Rscript --vanilla r/<module>.R params.json outdir`, see `METHODS.md`, section 1).
- The report is built by `utdata/blogg/build/build_rapport_unga.py` (`.venv/bin/python utdata/blogg/build/build_rapport_unga.py`), which computes the descriptive counts, reads the rest from the result files and stops if anything disagrees; the builder as reviewed before archiving is in commit `6eae0af`.
- In Vitki the association is shown under Statistics, The people, with the sensitivity analyses step by step. The report applies to the material's version `e83202e01a5e6d27`. The app's latest version (`3691a2c441dd03f4`) holds the Viking Age division constant in the sensitivity analyses (folklands, small lands and thirds, else the province) and so gives slightly different figures: adjusted odds ratio 3.77 instead of 3.90, with the carver as a random effect 4.09 instead of 4.41, and weakest without one area at a time 3.31 (without Östergötland) instead of 3.64 (without Uppland). Every step holds there too. Popular account: The journeys belonged to the young.

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Change log

- **Version 1, 2026-10-31.** First published version.
- **Version 2, 2026-11-05.** Corrections after a review before archiving in Zenodo. The main test, its figures and the title are unchanged.
- *Sons against fathers (5.2).* Version 1 gave the raw counts 26 of 230 against 13 of 496, counted with a wider definition than the model's (a son who was also a brother, a father who was also a husband). With the model's definition they are 26 of 204 against 12 of 449, 653 stones in all as in the model.
- *The wives (the abstract, 5.4, 6, the essentials box).* Version 1 said that only 4 of 69 stones for men who died abroad were raised by a wife and read it as most of them having no family of their own, without a comparison. Among the other stones for men the share is 117 of 1,148; the difference may be chance (p = 0.30).
- *What the model holds constant.* The model holds region constant, not province; the chart's heading, the abstract, 6 and 7 are corrected. ”In every province” is changed to ”without any one province”, since the check removes one province at a time.
- *Who travelled and who died (the abstract, 6, 7, the essentials box).* Version 1 said that the journeys abroad belonged chiefly to the young men. The stones measure death abroad, not who travelled; the text now says that death in the east and the west struck first of all men whom the stones remember as sons and brothers.
- *The hand coding of the outcome (3).* That it was partly done while the rules were being corrected, and that the rule was right on all 33 randomly drawn stones, is now stated.
- *Minor changes. drengr* ”appears on” the stone; Sukhino-Khomenko's full title; ”that according to Rundata stand in place”; the provinces behind the regions and Halland; the rule for the halves; simulated residuals and Fisher's test are explained; the chart's source line; Rundata's version and licence; absolute links; the commit of the analysis and the commands to rerun it; a link to the glossary; a consistent voice.
- **Version 3, 2026-11-05.** Corrections after a second, independent reading of the changes in version 2. The main test, its figures and the title are unchanged.
- *The essentials box.* Version 2 said that the result ”fits the journeys belonging to a stage of life before a family of one's own”. The measure does not say that, and the comparison of the wives (p = 0.30) gives it no support; the box now says, as section 6 does, that the stones cannot decide it.
- *Against earlier research (6).* ”The younger men of the family” is changed to ”men in a younger place in the family, sons and brothers”, since the measure is the family place, not age (weakness 1).
- *The hand coding of the outcome (3).* The text now says that 2 of the 33 randomly drawn stones say the dead man died abroad; that check thus mostly shows that the rule does not add deaths abroad.
- *Data and code.* The builder as reviewed before archiving is named with commit `6eae0af`.
- *What version 2 also changed without logging it.* Section 5.4 gained that parents or siblings stand among the sponsors on 46 of the 69 stones and children or a wife on 20. Data and code gained the app's latest figures (adjusted odds ratio 3.77, with the carver as a random effect 4.09, weakest 3.31 without Östergötland), and the sentence on a future DOI was removed.

Competing interests: none.
Funding: none.

Cite as

Kvant, V. (2026). Sons and brothers died outside Scandinavia more often than fathers and husbands – 10 against 3 percent. Research report, report version 3; analysis in Vitki 3.0.0. https://vitkirunor.se/en/forskningsrapporter/soner-och-broder-dog-utanfor-norden